

DBMS: Finding Conflict Equivalent Schedules

Comprehensive Quick-Revision Exam Notes & Master Cheat Sheet

1. What is Conflict Equivalence?

Given two schedules **S** and **S'**, they may look different in structure, but we aim to answer:

- Are **S** and **S'** conflict equivalent?
- Can we rearrange the operations of **S** to obtain **S'** without changing final outcomes?

Core Principle: Two schedules are conflict equivalent if one can be transformed into the other by swapping *adjacent non-conflicting operations*.

★ The Golden Rule for Conflict Pairs

Two operations conflict if and only if **ALL THREE** conditions are met:

Different Transactions + Same Data Item + At least One WRITE Operation

2. Conflict vs. Non-Conflict Pairs

Operation Pair	Same Data Item?	Contains Write?	Result Status
$R_1(A), R_2(A)$	Yes	No	✗ Non-Conflict (RR = Relax)
$R_1(A), W_2(A)$	Yes	Yes	● Conflict (RW)
$W_1(A), R_2(A)$	Yes	Yes	● Conflict (WR)
$W_1(A), W_2(A)$	Yes	Yes	● Conflict (WW)
$R_1(A), W_2(B)$	No ($A \neq B$)	Yes	✗ Non-Conflict
$W_1(A), R_2(B)$	No ($A \neq B$)	Yes	✗ Non-Conflict

3. Operation Swapping Rules

✓ Non-Conflicting Pair

Adjacent operations operating on different items or both reading can be swapped freely.

$W_1(A) \cdot R_2(B) \rightarrow R_2(B) \cdot W_1(A)$

✗ Conflicting Pair

Operations conflicting on the same item CANNOT be interchanged.

$W_1(A) \cdot R_2(A) \rightarrow \text{Cannot Swap!}$

★ Important Constraint: Operations Are Preserved

Swapping only modifies the **relative positions** of operations. It NEVER alters the transaction ID, operation type (READ/WRITE), or data item target.

4. Step-by-Step Method to Find Equivalent Schedules

- 1 **Compare Schedules:** Position the given schedule S and required schedule S' side by side.
- 2 **Identify Target Order:** Identify adjacent operations in S that need reordering to match S'.
- 3 **Check Conflict Conditions:** Evaluate if the adjacent pair is conflicting or non-conflicting.
- 4 **Perform Permissible Swaps:** Swap only if the adjacent pair is non-conflicting.
- 5 **Iterate:** Repeat step-by-step swapping until target schedule S' is obtained or blocked by a conflict.

5. Walkthrough Examples

Example A: Valid Swap Allowed

Goal: Move $R_2(B)$ above $W_1(A)$.

Check: $W_1(A)$ & $R_2(B)$

- Different transactions: T_1, T_2 ✓
- Different items: $A \neq B$ ✓

Result: Non-Conflict → SWAP ALLOWED ✓

Example B: Invalid Swap Blocked

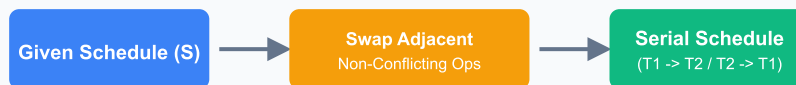
Goal: Move $R_2(A)$ above $W_1(A)$.

Check: $W_1(A)$ & $R_2(A)$

- Different transactions: T_1, T_2 ✓
- Same item: A ✓ | Contains WRITE ✓

Result: Conflict → SWAP BLOCKED ✗

6. Relationship with Conflict Serializability



Conclusion: If a schedule can be swapped into a serial schedule, it is Conflict Serializable.

7. ⚠️ High-Yield Exam Traps & Quick Quiz

Common Pitfall Questions:

- Q: Are $R_1(A)$ and $W_2(B)$ conflicting?
→ ✗ No! Different data items ($A \neq B$).
- Q: Are $R_1(A)$ and $R_2(A)$ conflicting?
→ ✗ No! Both are READ operations.
- Q: Are $W_1(A)$ and $W_2(A)$ conflicting?
→ ✓ Yes! Same data item and both are WRITE operations.
- Q: Do operations within the SAME transaction conflict?
→ ✗ Not for equivalence swapping rules! Conflict pairs are evaluated across *different* transactions.

8. 🏆 One-Minute Exam Revision Summary

- RR = Relax (No conflict) | RW / WR / WW = Warning (Conflict if same data item).
- Non-conflict pair → Swap allowed ($AB \rightarrow BA$) ✓
- Conflict pair → Swap forbidden ($AB \rightarrow BA$) ✗
- Final Formula: Different Transactions + Same Data Item + At Least One Write = **Conflict**